

Control work 8, and work on the mistakes

Algebraic fractions from end to end — cancelling, common denominators, products and quotients.

LESSON 120–121

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 8

S8 7 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Cancel only after factorising, and state the excluded values.
- Build the lowest common denominator and add or subtract correctly.
- Multiply and divide fractions, cancelling first.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 327–357
Cambridge	Stage 8, pp. 66–78
Workbook	Control paper M8

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Simplify $\frac{x^2 - 25}{x + 5}$ and state the excluded value	4	L112–113
2	Find the excluded values of $\frac{3}{x^2 - 4x}$	3	L112–113
3	Find the lowest common denominator of $x^2 - 9$ and $x + 3$	3	L114–115
4	Simplify $\frac{1}{x - 2} + \frac{3}{x + 2}$	5	L116–117
5	Simplify $\frac{4x + 1}{x - 3} - \frac{2x + 7}{x - 3}$	4	L116–117
6	Simplify $\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$	6	L118–119

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for $x \neq -5$ alone; Q4 two for the extra factors and one for expanding the numerators; Q5 two for the bracket; Q6 two for inverting the right fraction and one for the hidden exclusion $x \neq -1$.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
terms cancelled	$\frac{x^2 - 25}{x + 5} = x^2 - 5$	factorise, then cancel factors
excluded value forgotten	an answer with no condition	$x \neq -5$
only one excluded value found	$x \neq 0$	$x \neq 0$ and $x \neq 4$
denominators multiplied blindly	$(x - 3)(x + 3)^2$	$(x - 3)(x + 3)$
numerator not multiplied up	$\frac{1 + 3}{x^2 - 4}$	$\frac{(x + 2) + 3(x - 2)}{x^2 - 4}$
bracket missing on subtraction	$4x + 1 - 2x + 7$	$(4x + 1) - (2x + 7)$
wrong fraction inverted	$\frac{x}{x^2 - 1} \cdot \frac{x + 1}{x^2}$	invert the divisor

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

OPERATION	WHAT IT NEEDS	WHAT IT DOES NOT NEED
cancelling	both parts factorised	a common denominator
adding	the LCD and brackets	anything inverted
subtracting	the LCD and a bracket on the second numerator	—
multiplying	factorising, then cancelling	a common denominator
dividing	inverting the divisor	a common denominator

LOOKING FORWARD

The last block of the quarter is equations: what a root is, how a linear equation is solved, Al-Khwarizmi's method, and problems written as equations. Fractions will appear inside those equations, so this chapter stays in use.

02 Worked examples

EXAMPLE 1 Model answer, Q2: the excluded values of $\frac{3}{x^2 - 4x}$.

① $x^2 - 4x = x(x - 4)$
Factorise the denominator.

② It is zero when $x = 0$ or $x = 4$.

③ $x \neq 0$ and $x \neq 4$.
Both, not one.

ANSWER

$x \neq 0, x \neq 4$

EXAMPLE 2 Model answer, Q4: $\frac{1}{x-2} + \frac{3}{x+2}$.

① $\text{LCD} = (x-2)(x+2) = x^2 - 4.$

② $\frac{x+2}{x^2-4} + \frac{3x-6}{x^2-4}$

Numerators multiplied out.

③ $= \frac{4x-4}{x^2-4}$

④ $= \frac{4(x-1)}{(x-2)(x+2)}, x \neq \pm 2.$

Nothing cancels.

ANSWER

$$\frac{4x-4}{x^2-4}$$

EXAMPLE 3 Model answer, Q6: $\frac{x^2-1}{x} \div \frac{x+1}{x^2}$.

① Invert the divisor: $\frac{x^2-1}{x} \cdot \frac{x^2}{x+1}$.

② $x^2 - 1 = (x-1)(x+1)$

③ Cancel $x+1$ and one x .

④ $= x(x-1)$, with $x \neq 0$ and $x \neq -1$.

The second one is hidden.

ANSWER

$$x(x-1)$$

03 Interactive model

Mark Q5 for the bracket alone and Q6 for the exclusions alone; the class sees that the algebra was rarely the problem.

The chapter in twelve questions

An algebraic fraction has no value when:

the numerator is zero

the denominator is zero

both are zero

never

$\frac{x^2 - 25}{x + 5}$ equals:

$x^2 - 5$

$x - 5$

$x + 5$

x

Its excluded value is:

$x \neq 5$

$x \neq -5$

$x \neq 0$

none

3 $x^2 - 4x$ excludes:

$x \neq 0$

$x \neq 4$

both

neither

The LCD of $x^2 - 9$ and $x + 3$ is:

$(x - 3)(x + 3)^2$

$x^2 - 9$

$x + 3$

$(x + 3)^2$

Cancelling is allowed for:

terms

factors

both

neither

When subtracting, the second numerator needs:

a bracket

a denominator

a factor

nothing

$(4x + 1) - (2x + 7)$ equals:

To divide by a fraction you invert:

$\frac{x^2 - 1}{x} \div \frac{x + 1}{x^2}$ equals:

\frac{x - 1}{x}"/>

\frac{1}{x}"/>

Its hidden exclusion is:

$$x \neq 1$$

$$x \neq -1$$

$$x \neq 2$$

none

In a product you should cancel:

after expanding

before expanding

never

only numbers

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
To cancel	Qisqartirish	Сокращать
Common denominator	Umumiy maxraj	Общий знаменатель
Reciprocal	Teskari kasr	Обратная дробь
Bracket	Qavs	Скобка
Excluded value	Mumkin bo'lmagan qiymat	Недопустимое значение
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 begins by:

cancelling

factorising

expanding

substituting

Q2 needs how many excluded values?

0

1

2

3

Q3 is answered with:

 $(x - 3)(x + 3)^2$ $x^2 - 9$ $x + 3$ $(x - 3)^2$

Q4 loses marks most often at:

the LCD

multiplying the numerators up

the final answer

the exclusions

Q5 is about:

the LCD

the bracket

the reciprocal

the exclusions

Q6 requires inverting:

the first fraction

the second fraction

both

neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.
$$\frac{x^2 - 25}{x + 5}$$

ANSWER $x - 5$

2. Its excluded value

ANSWER $x \neq -5$

3.
$$\frac{3}{x^2 - 4x}$$
: the excluded values

ANSWER $x \neq 0, x \neq 4$

4. LCD of $x^2 - 9$ and $x + 3$

ANSWER $x^2 - 9$

5. $(4x + 1) - (2x + 7)$

ANSWER $2x - 6$

6.
$$\frac{3}{x} + \frac{5}{x}$$

ANSWER $\frac{8}{x}$

7.
$$\frac{2}{x} \cdot \frac{x}{4}$$

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\frac{1}{x-2} + \frac{3}{x+2}$

ANSWER $\frac{4x-4}{x^2-4}$

2. $\frac{4x+1}{x-3} - \frac{2x+7}{x-3}$

ANSWER $\frac{2x-6}{x-3} = 2$

3. $\frac{x^2-1}{x} \div \frac{x+1}{x^2}$

ANSWER $x(x-1)$

4. Its excluded values

ANSWER $x \neq 0, x \neq -1$

5. $\frac{x^2+6x+9}{x+3}$

ANSWER $x+3$

6. LCD of x^2-4 and $x+2$

ANSWER x^2-4

7. $\frac{x}{x+1} \cdot \frac{x^2-1}{x}$

ANSWER $x-1$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{x-1} - \frac{2}{x^2-1}$

ANSWER $\frac{1}{x+1}$

2. $\frac{x^2-4}{x^2-4x+4}$

ANSWER $\frac{x+2}{x-2}$

3. $\frac{a}{a-b} - \frac{b}{a-b}$

ANSWER 1

4. $\frac{x^2-9}{x+2} \div \frac{x-3}{x+2}$

ANSWER $x+3$

5. $\frac{1}{x} + \frac{1}{x+1}$ as one fraction

ANSWER $\frac{2x+1}{x(x+1)}$

6. Why does Q6 exclude $x = -1$?

ANSWER That numerator becomes a denominator on inverting

7. $\frac{x^3+8}{x+2}$

ANSWER $x^2 - 2x + 4$

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.

2. Simplify $\frac{x^2 - 36}{x - 6}$ and state the excluded value.
3. Simplify $\frac{2}{x - 1} + \frac{1}{x + 1}$.
4. Simplify $\frac{x^2 - 4}{x} \div \frac{x - 2}{x^2}$ and list every excluded value.
5. Write one sentence for each row of the chapter map.